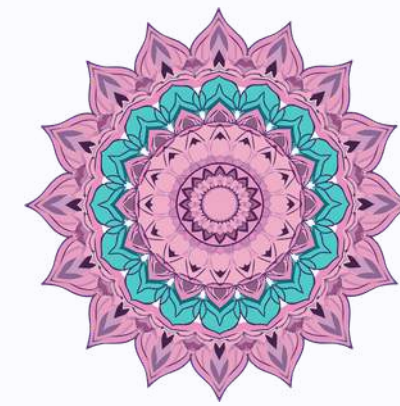




RECURSION:

**THE INFINITE
MIRROR
OF LOGIC**





The Dream Within a Dream

A character wakes up, only to find they are still dreaming — like layers within layers.

"Are we awake? Or dreaming again?"

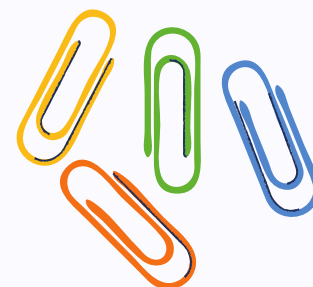




What is Recursion? "Small Roop"

Recursion means a function calls itself — like looking into two mirrors facing each other!

Key Term: "Small Roop" means Smaller Version — each call creates a smaller version of the same problem.

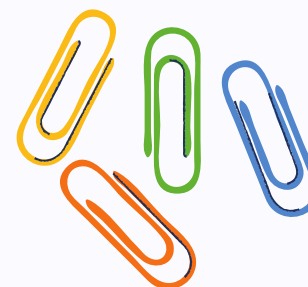
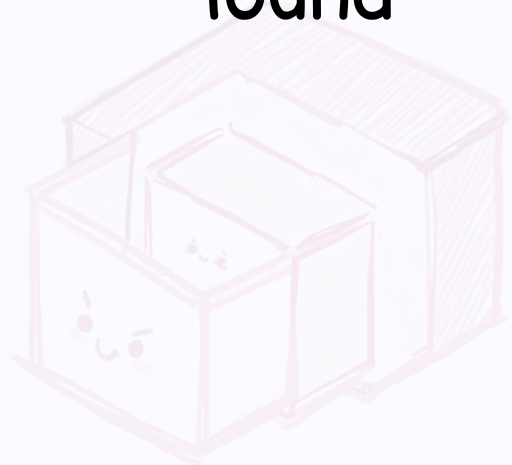




The Recursive Loop

Imagine opening a big dabba to find a smaller dabba inside, then a smaller one again, until the tiniest dabba holds the treat! This is exactly how recursion works — each step reveals a smaller version of the same problem.

- ✔ Problem broken into smaller same problems ("Small Roop")
- ✔ Process repeats until smallest dabba found

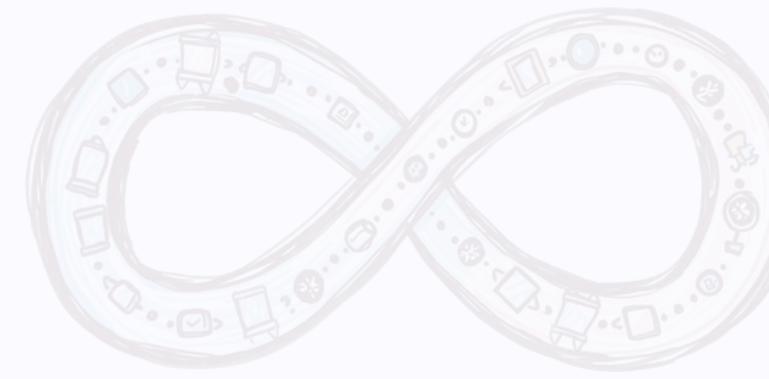




The Barber Shop Mirror

In a barber shop, the front mirror reflects your image, which reflects the back mirror, and so on — an infinite line of you!

This is called "Anant Pratibimb" — Infinite Reflection. It's the perfect example of recursion without a base case!

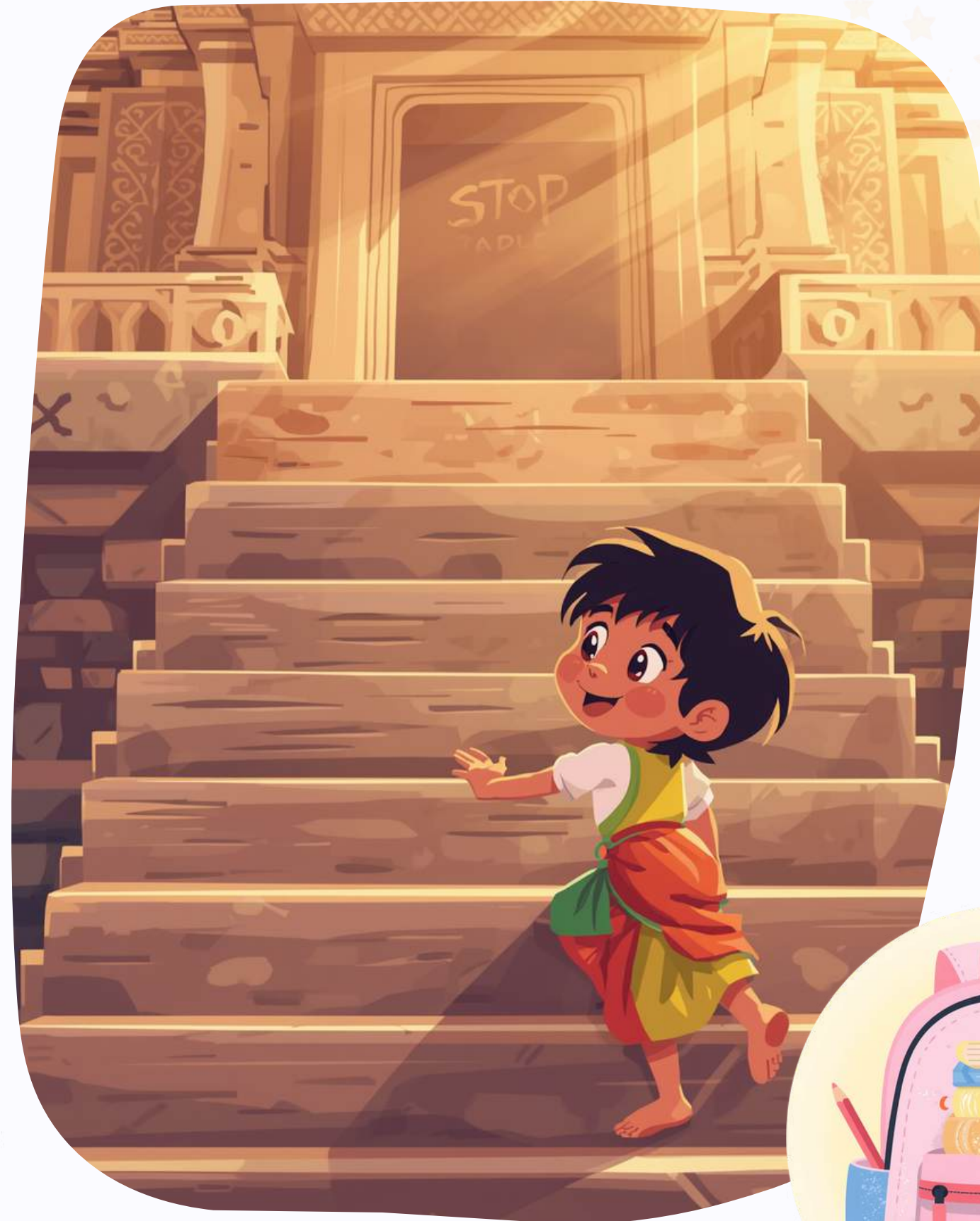




The Base Case: The Stop Rule

The base case tells recursion when to stop, like the last step before the temple door — the "Aakhri Padav." Without it, recursion would go on forever!

- ✓ If at door → Stop climbing
- ✓ Else → Climb the next step





The Recursive Case

The Repeat

Rule

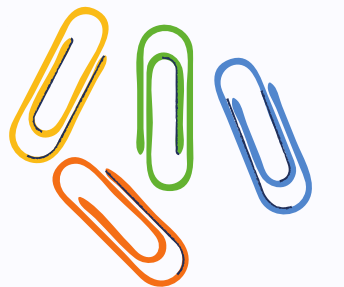
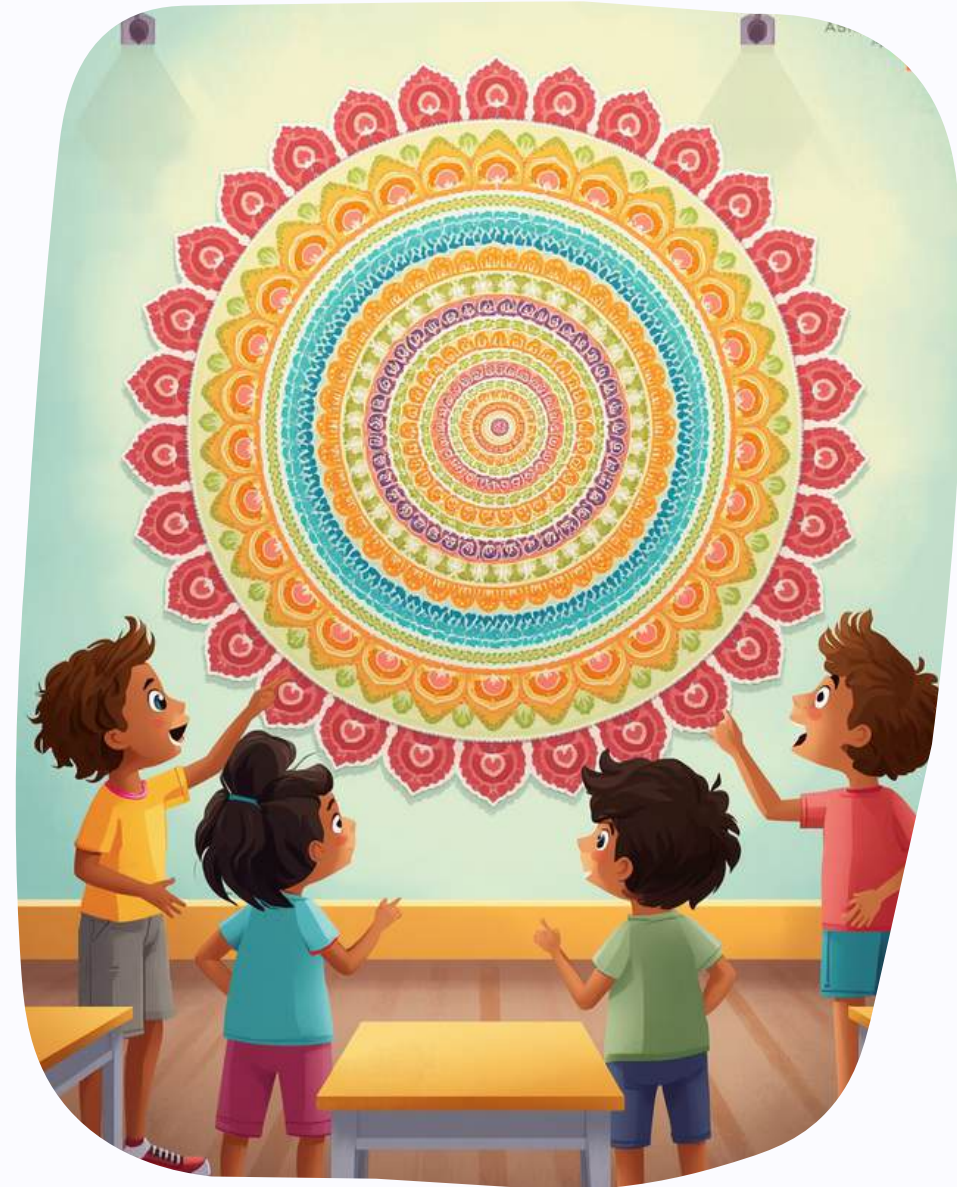
The recursive case says: "Do the same action again, but smaller or closer to the base case."

Example: "Climb one step, then climb the next step..." Each action repeats until you reach the Aakhri Padav (base case).





Recursion in Nature & Art

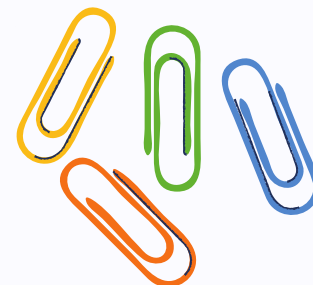
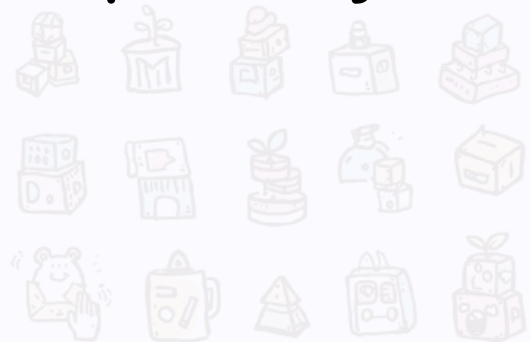




Understanding the Stack: The Waiting Line

Each step waits for the smaller step to finish — like a queue of friends waiting their turn to pass a message or baton.

This waiting is called the 'Stack' in recursion. Just like friends in line, each step must wait patiently until the one ahead finishes!





Interactive Activity

The Human Recursive Chain: Pass a message that gets shorter every time until it disappears — just like recursion! This fun game helps us understand how recursive functions work step by step.

- ✔ Start with a full sentence, each student repeats with one word less.
- ✔ Stop when message is empty — that's your Base Case (Aakhri Padav)!

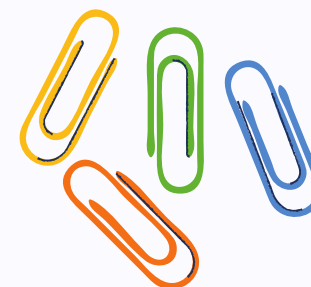




What Did We Learn?

We saw how each step depends on the next smaller step, and when to stop. The activity showed us recursion in action — breaking down a problem until we reach the end!

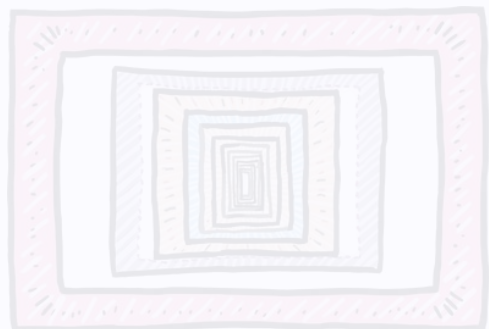
- ✔ Messages got smaller each time → Recursive Case (Small Roop)
- ✔ When message ended → Base Case (Aakhri Padav)





One Sentence Recap

Recursion is when a function calls itself with a smaller problem, and stops at the base case — the "Aakhri Padav".





Key Takeaways



Chhota Roop

Recursion breaks big problems into smaller same problems — like nested dabbas!

Aakhri Padav

The Base Case stops infinite loops — like reaching the temple door!

The Stack

Each step waits for smaller steps to finish — like friends in a queue!





THANK YOU!

Any Questions?

Recursion: The Infinite Mirror of Logic

